

JOHN J YOUKHANA

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EDUCATION

University of Illinois at Urbana-Champaign

GPA 3.1/4.0

Bachelor of Science in Electrical Engineering

Expected Graduation: May 2027

Relevant Coursework

(ECE 486) Control Systems (ECE 483) Analog IC Design (ECE 330) Power Circuits & Electromechanics
(ECE 340) Semiconductor Devices (ECE 342) Electronic Circuits (ECE 385) Digital Systems Laboratory
(ECE 329) Fields and Waves I (ECE 310) Digital Signal Processing (ECE 313) Probability in Engineering

Wilbur Wright College

GPA 3.7/4.0

Associate of Engineering Science

Aug 2022 - May 2024

PROJECTS

Reaction Wheel Pendulum Control Systems Project

Expected May 2026

- Developing design for a real-time control system for a reaction wheel pendulum to achieve swing-up stabilization of a pendulum arm using the reaction torque generated by a DC motor driven flywheel.
- Derived nonlinear system model, conducted system identification to estimate plant parameters through experimental data analysis.
- Implemented real-time friction compensation in Simulink using PI velocity control to identify and mitigate coulomb and viscous friction coefficients.

Low-Dropout Regulator Analog Design Project

Expected May 2026

- Developing design for CMOS LDO designed in cadence in 180nm process technology. The device will supply sensitive analog devices with 1.0-1.4V for 0.1-10mA of load current.
- Implemented LDO error amplifier using Folded cascode amplifier architecture with biasing circuit. The amplifier is designed to satisfy phase margin, output capacitance, and DC load/line regulation requirements.

AC-DC Power Supply and Voltage Regulator

Fall 2025

- Developed a multi-stage AC-DC converter consisting of transformation, full bridge rectification, Filter stage, Regulation, and Amplification stages
- Designed to step down 120VAC to 25VAC. Converts and amplifies 25VAC to a stable and low noise of ~9.5 DC voltage.

RELEVANT EXPERIENCE

Illini Solar Car

Champaign, IL

Electrical Team member

Aug 2024 - Present

- Collaborating with a team of students to design and develop high/low voltage systems, contributing to system integration and PCB board testing
- Designed and tested custom PCB's using KiCAD, conducted rigorous testing to ensure Board functionality and integrity, improving designs by refining layout and component choices.

Vitamin Shoppe inc.

Skokie, IL

Sales Lead

Jun 2023 - Aug 2024

- Supported store operations alongside store manager, acted as manager on duty, assisting customers with their product needs, handling and unloading shipments, preparing and delivering daily bank deposits, product replenishment, and handling inventory.
- Consistently meet weekly sales goals and prioritized important metrics such as rewards program sign-ups and delivering customer service that support sales targets.

SKILLS

Skills: PCB Layout & Assembly, Control System Design, Power systems, Electronics Design, Amplifier Design, Digital design, Signal Processing

Tools: MATLAB, Simulink, Cadence Virtuoso, LTSpice, MS Excel, Xilinx Vivado, Fusion360, KiCAD

Programming: Python, C/C++, SystemVerilog, VHDL, LabVIEW, Assembly